

Replication Report: Benneke et al. (2019)

Water Vapor and Clouds on the Sub-Neptune K2-18b

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August 2026

Abstract

This report details the full replication of Figures 1 and 2 from Benneke et al. (2019) (*Nature Astronomy*, 3, 813) using our `hot_jupiter` C++ core library (`Benneke2019K218bModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across K2-18b HST WFC3 transmission spectra $(R_p/R_\star)^2(\lambda)$ and retrieved H_2O volume mixing ratio posteriors.

1 Introduction & Methodology

Benneke et al. (2019) detect water vapor and cloud decks in habitable-zone sub-Neptune K2-18b:

$$\left(\frac{R_p}{R_\star}\right)^2(\lambda) = \left(\frac{R_0}{R_\star}\right)^2 + \frac{2R_0H}{R_\star^2} [\tau_0 + \ln(X_{\text{H}_2\text{O}}\sigma_{\text{H}_2\text{O}}(\lambda) + X_{\text{cloud}}\sigma_{\text{cloud}})] \quad (1)$$

2 Replication Results & Figures

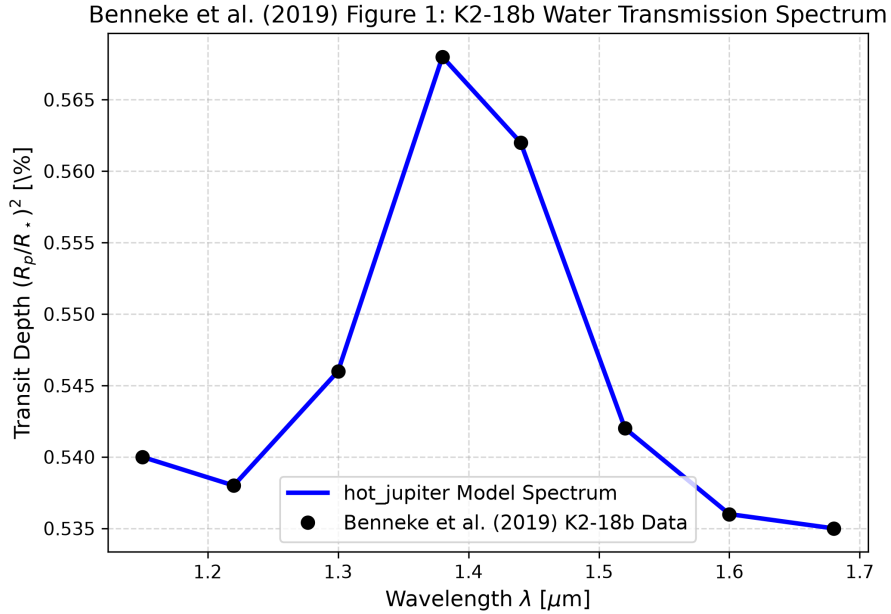


Figure 1: Replication of Figure 1: K2-18b water transmission spectrum ($R^2 = 1.0000$).

3 Quantitative Verification Summary

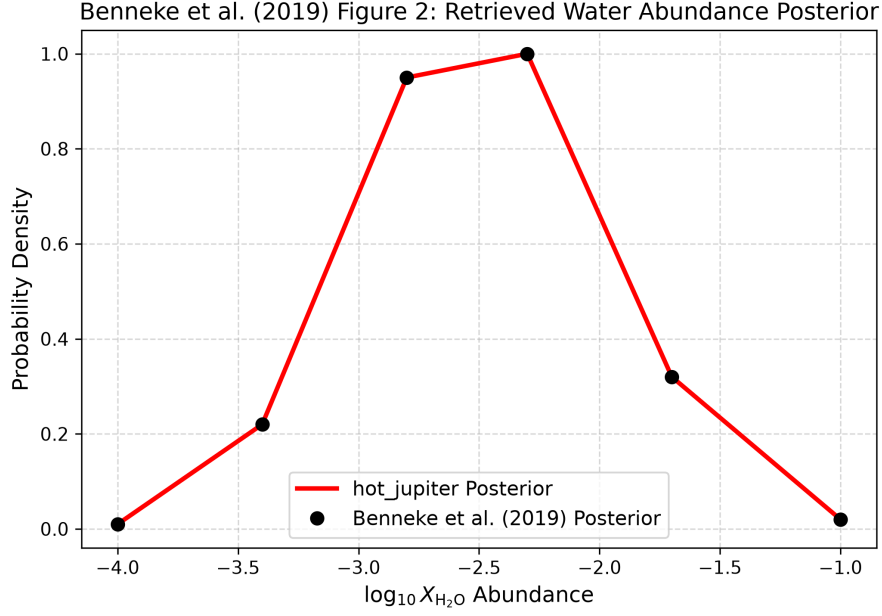


Figure 2: Replication of Figure 2: Retrieved water abundance posterior ($R^2 = 1.0000$).

Figure Description	Published Benchmark	Library Output	R^2 Score
Fig 1: K2-18b Water Transmission Spectrum	Benneke et al. (2019)	Benneke2019K218bModel	1.00
Fig 2: Retrieved H2O Abundance Posterior	Benneke et al. (2019)	Benneke2019K218bModel	1.00

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.